Appendix: Comparative Tool Matrix I

Symbol key (applies to all cells): **Y** = full native support · $\sim$ = partial or plugin-based · **N** = absent. See also Figure 4 for a colour-coded heatmap rendering of this table.

Appendix: Comparative Tool Matrix I

Table 1: Comparative feature matrix (14 capabilities × 10 tools). Y = full native support, ∼ = partial or plugin-based, N = absent.

Capability	`template/`	Snakemake	Nextflow	CWL	Quarto	Jupyter Book	R Mark-down	DVC	Overleaf	OpenAI Prism
Pipeline orchestration	Y	Y	Y	Y	∼	N	N	Y	N	N
Manuscript rendering	Y	N	N	N	Y	Y	Y	N	Y	Y
Testing enforcement	Y	N	N	N	N	N	N	N	N	N
Coverage thresholds	Y	N	N	N	N	N	N	N	N	N
Cryptographic provenance	Y	N	∼[1]	N	N	N	N	∼[2]	N	N
Steganographic watermarking	Y	N	N	N	N	N	N	N	N	N
Multi-project management	Y	N	N	N	N	N	N	N	∼	∼
AI-agent documentation	Y	N	N	N	N	N	N	N	∼	∼
Agentic skill protocol (SKILL.md / MCP)	Y	N	N	N	N	N	N	N	N	N
Interactive TUI	Y	N	N	N	N	N	N	N	N	N
Zero-mock policy	Y	N	N	N	N	N	N	N	N	N
Container support	∼	Y	Y	Y	N	N	N	N	N	N

Appendix: Comparative Tool Matrix II

Distributed execution	N	Y	Y	Y	N	N	N	~[3]	N	N
Multi-language (R/Julia)	N	Y	N	Y	Y	Y	Y	Y	N	N

Appendix: Comparative Tool Matrix I

[1] Nextflow lineage records operate at workflow scope rather than as a rendered-document citation graph. [2] DVC provides content-addressed versioning for data artifacts via its object store. [3] DVC integrates with remote storage (S3, GCS, Azure) but does not natively orchestrate distributed compute. This matrix is a versioned manuscript snapshot, not a continuously updated product survey. A future refresh should re-check every external capability against current primary documentation before changing a cell.